

1. Lab 13: Transport Layer Security (TLS), PKI, and Routed DNS Infrastructure Setup

Lab Name: Lab 13: Transport Layer Security (TLS), PKI, and Routed DNS Infrastructure Setup **Date:** YYYY-MM-DD
Name: [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML), Cisco IOSv / Linux Router, OpenSSL, dnsmasq , Nginx, curl , Wireshark

AI Policy & Acceptable Use Guidelines



AI Policy & Acceptable Use Guidelines - Yellow Icon

Artificial Intelligence tools can be powerful learning aids, but they must be used ethically and responsibly to ensure you develop core networking skills.

- **Lab Assignments:** Do not use AI to perform lab steps or write your lab submissions. However, if you get stuck on a concept or technical obstacle, you may use AI as a virtual tutor to ask clarifying questions and help you troubleshoot.
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2. Background

This capstone lab brings together a comprehensive synthesis of the core networking and security principles you have mastered throughout the semester. In previous labs, you explored isolated protocol layers—such as IP addressing and subnetting, inter-subnet routing, DNS domain resolution, TCP connection management, and link-layer bridging. In this lab, you will combine all of these elements into a single, fully integrated end-to-end secure web infrastructure.

Modern secure web communications rely on the seamless integration of four fundamental network technologies:

1. **Inter-Subnet IP Routing:** Routers forward IP packets across distinct network subnets (e.g., Client LAN to Server DMZ) using default gateways and routing tables.
2. **Domain Name System (DNS):** Translates human-readable domain names (e.g., <lastname>.cs457.lab) into IP addresses (10.1.2.100) over UDP Port 53.
3. **Public Key Infrastructure (PKI):** Establishes digital identity and trust. A **Root Certification Authority (CA)** signs digital certificates (server.crt), binding a web server's identity to its public key.
4. **Transport Layer Security (TLS):** Operating over TCP Port 443, TLS uses asymmetric cryptography for server authentication and session key exchange, followed by symmetric key encryption (e.g., AES) for bulk application data (HTTPS).

When a user visits an HTTPS web page:

- The client queries its configured **DNS Server** to resolve the server's hostname to an IP address.
- The client establishes a **TCP 3-Way Handshake** across intermediate **Routers** to the destination web server.
- The client performs a **TLS Handshake**, receiving the server's X.509 certificate.
- If the client trusts the **Root CA** that signed the certificate, the client validates the identity without warnings and establishes an encrypted HTTPS session.

In this lab, you will construct and configure a multi-subnet CML topology featuring an Edge Router, a DNS Server, a Web Server running a Private PKI, and a Web Client, analyzing the complete multi-layer protocol stack in Wireshark.

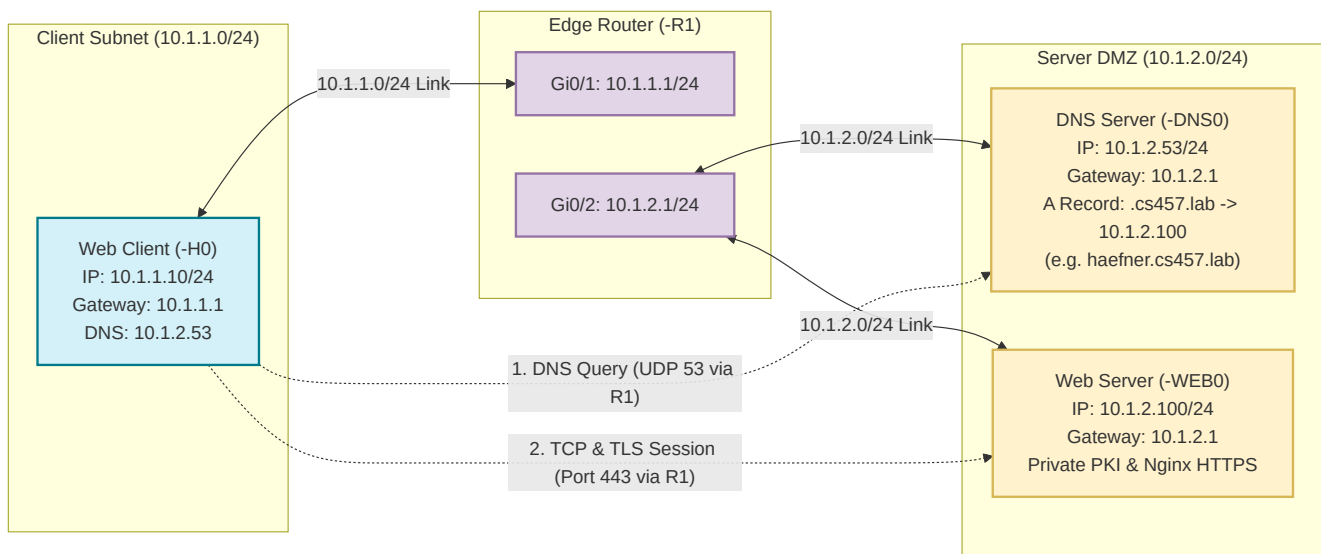
3. Objective / Purpose

To configure an inter-subnet router, set up a DNS server for custom domain resolution, build a private PKI to issue X.509 certificates, configure an HTTPS web server, push the Root CA certificate to the client, and capture/analyze the complete DNS, TCP, and TLS protocol flow in Wireshark.

4. Topology Diagram

[!IMPORTANT] Custom Student Domain Naming Rule: Every student **MUST** use their own lower-case last name for the domain name: `<lastname>.cs457.lab`.

- **Example (Prof. Haefner):** `haefner.cs457.lab`
- Replace `<lastname>` with your actual last name in all DNS configurations, OpenSSL certificate requests, Nginx server blocks, `nslookup`, and `curl` tests!



Details:

- Client Subnet: 10.1.1.0/24 (Gateway: 10.1.1.1)
- Server Subnet: 10.1.2.0/24 (Gateway: 10.1.2.1)
- Router (<initials>-R1): Dual-homed routing between Client LAN and Server DMZ
- Services: DNS (10.1.2.53), HTTPS (10.1.2.100 FQDN: <lastname>.cs457.lab , e.g. haefner.cs457.lab)

5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Cisco IOSv / Linux Router, OpenSSL, dnsmasq, Nginx, curl, Wireshark.
- **Nodes:** 1x Edge Router (<initials>-R1), 1x Linux Client (<initials>-H0), 1x Linux DNS Server (<initials>-DNS0), 1x Linux Web Server (<initials>-WEB0).

6. Step-by-Step Configuration Guide

[!IMPORTANT] Domain Name Customization: Throughout these steps, replace `<lastname>` with your lower-case last name (e.g., `haefner.cs457.lab`).

Step 1: Configure Inter-Subnet Router (`<initials>-R1`)

Open the console of Router `<initials>-R1` and configure interface IP addresses and IP routing:

- **Cisco IOSv Router Configuration:**

```
enable
configure terminal
hostname KH-R1

! Client Subnet Interface
interface GigabitEthernet0/1
description Client-LAN-Gateway
ip address 10.1.1.1 255.255.255.0
no shutdown

! Server DMZ Interface
interface GigabitEthernet0/2
description Server-DMZ-Gateway
ip address 10.1.2.1 255.255.255.0
no shutdown

end
write memory
```

(Alternative for Linux Router node: `ip addr add 10.1.1.1/24 dev eth1 && ip addr add 10.1.2.1/24 dev eth2 && sysctl -w net.ipv4.ip_forward=1`).

Step 2: Configure Node IP Addressing & Default Gateways

1. **Client Node (`<initials>-H0`):**

```
# Assign IP and default gateway pointing to R1 Gi0/1
ip addr add 10.1.1.10/24 dev eth0 2>/dev/null || ip addr add 10.1.1.10/24 dev wlan0
ip link set eth0 up 2>/dev/null || ip link set wlan0 up
ip route add default via 10.1.1.1
```

2. **DNS Server Node (`<initials>-DNS0`):**

```
# Assign IP and default gateway pointing to R1 Gi0/2
ip addr add 10.1.2.53/24 dev eth0 2>/dev/null || ip addr add 10.1.2.53/24 dev wlan0
ip link set eth0 up 2>/dev/null || ip link set wlan0 up
ip route add default via 10.1.2.1
```

3. **Web Server Node (`<initials>-WEB0`):**

```
# Assign IP and default gateway pointing to R1 Gi0/2
ip addr add 10.1.2.100/24 dev eth0 2>/dev/null || ip addr add 10.1.2.100/24 dev wlan0
ip link set eth0 up 2>/dev/null || ip link set wlan0 up
ip route add default via 10.1.2.1
```

4. **Verify Routing Reachability:** From `<initials>-H0` , ping across the router to both servers:

```
ping -c 2 10.1.2.53
ping -c 2 10.1.2.100
```

Step 3: Configure DNS Server (`<initials>-DNS0`)

On `<initials>-DNS0` , configure `dnsmasq` to resolve `<lastname>.cs457.lab` (e.g. `haefner.cs457.lab`) to `10.1.2.100` :

```
# Start dnsmasq resolving <lastname>.cs457.lab -> 10.1.2.100
# Example for Prof. Haefner: dnsmasq -k --address=/haefner.cs457.lab/10.1.2.100 --listen-
address=10.1.2.53 &
pkill dnsmasq 2>/dev/null
dnsmasq -k --address=/<lastname>.cs457.lab/10.1.2.100 --listen-address=10.1.2.53 &
```

On `<initials>-H0` , point local DNS resolution to `<initials>-DNS0` :

```
echo "nameserver 10.1.2.53" > /etc/resolv.conf

# Test DNS resolution across the router (e.g. nslookup haefner.cs457.lab 10.1.2.53)
nslookup <lastname>.cs457.lab 10.1.2.53
```

Step 4: Build Private PKI & Issue Server Certificate

On the Web Server node (`<initials>-WEB0`), create the private Root CA and sign the web server certificate using OpenSSL:

```
mkdir -p /etc/ssl/pki
cd /etc/ssl/pki

# 1. Generate Private Key & Self-Signed Root CA Certificate
openssl genrsa -out ca.key 2048
openssl req -x509 -new -nodes -key ca.key -sha256 -days 365 \
-out ca.crt \
-subj "/C=US/ST=Colorado/L=FortCollins/O=CS457-PKI/CN=CS457-Root-CA"

# 2. Generate Private Key & SAN Configuration for Web Server
openssl genrsa -out server.key 2048

# NOTE: Replace <lastname> below with your lower-case last name (e.g. haefner.cs457.lab)
cat << 'EOF' > san.cnf
[req]
```

```

distinguished_name = req_distinguished_name
req_extensions = v3_req
prompt = no

[req_distinguished_name]
C = US
ST = Colorado
L = FortCollins
O = CS457-Lab
CN = <lastname>.cs457.lab

[v3_req]
keyUsage = keyEncipherment, dataEncipherment
extendedKeyUsage = serverAuth
subjectAltName = @alt_names

[alt_names]
DNS.1 = <lastname>.cs457.lab
IP.1 = 10.1.2.100
EOF

# 3. Generate CSR & Sign Server Certificate with Root CA
openssl req -new -key server.key -out server.csr -config san.cnf
openssl x509 -req -in server.csr \
    -CA ca.crt -CAkey ca.key -CAcreateserial \
    -out server.crt -days 365 -sha256 -extfile san.cnf -extensions v3_req

```

Step 5: Configure HTTPS Web Server

On **<initials>-WEB0**, configure Nginx to serve HTTPS traffic over Port 443:

```

mkdir -p /var/www/html
cat << 'EOF' > /var/www/html/index.html
<!DOCTYPE html>
<html>
<head><title>CS457 Secure Web Server</title></head>
<body>
    <h1>Welcome to CS457 Routed TLS & PKI Web Server</h1>
    <p>This page was resolved via DNS (10.1.2.53) and encrypted via TLS (10.1.2.100)!</p>
</body>
</html>
EOF

# NOTE: Replace <lastname> with your lower-case last name (e.g. haefner.cs457.lab)
cat << 'EOF' > /etc/nginx/sites-available/default
server {
    listen 443 ssl;
    server_name <lastname>.cs457.lab;

    ssl_certificate /etc/ssl/pki/server.crt;
    ssl_certificate_key /etc/ssl/pki/server.key;

```

```
ssl_protocols TLSv1.2 TLSv1.3;
ssl_ciphers HIGH:!aNULL:!MD5;

location / {
    root /var/www/html;
    index index.html;
}
}
EOF

nginx -t && nginx -s reload 2>/dev/null || service nginx restart
```

Step 6: Push Root of Trust (ca.crt) to Client & Verify

1. **Transfer Root CA Certificate to Client (<initials>-H0):** On <initials>-WEB0 , display ca.crt :

```
cat /etc/ssl/pki/ca.crt
```

On <initials>-H0 , save ca.crt :

```
cat << 'EOF' > /tmp/ca.crt
[PASTE ca.crt CONTENT HERE]
EOF
```

2. **Install Root CA into Client Trust Store:** On <initials>-H0 :

```
cp /tmp/ca.crt /usr/local/share/ca-certificates/cs457-root-ca.crt 2>/dev/null || cp
/tmp/ca.crt /etc/ssl/certs/cs457-root-ca.crt
update-ca-certificates 2>/dev/null
```

3. **Verify Untrusted vs. Trusted TLS Connection:** On <initials>-H0 , test fetching the web page **without** specifying the CA certificate (fails due to untrusted CA):

```
# Example: curl -v https://haefner.cs457.lab
curl -v https://<lastname>.cs457.lab
```

Now test **with** the Root of Trust passed explicitly:

```
# Example: curl -v --cacert /tmp/ca.crt https://haefner.cs457.lab
curl -v --cacert /tmp/ca.crt https://<lastname>.cs457.lab
```

(Verify that output prints `SSL certificate verify ok.` and displays the encrypted HTML page!).

Step-by-Step Traffic Generation & Wireshark Capture:

[!IMPORTANT] **REMINDER: All screenshots, console outputs, and packet captures (.pcap files) must be taken from the SAME active lab session. Mixing screenshots or trace files across different runs or**

sessions is not permitted.

1. **Start Packet Capture:** In CML GUI, right-click the Client (<initials>-H0) interface or Router (<initials>-R1) interface Gi0/1 , select **Packet Capture**, and click **Start**.
2. **Generate Traffic:** On Client <initials>-H0 , execute:

```
curl -v --cacert /tmp/ca.crt https://<lastname>.cs457.lab
```

3. **Stop & Download Capture:** In CML GUI Packet Capture tab, click **Stop** and **Download** the .pcap file to open in Wireshark.

Extra Credit Challenge (+10 Points Bonus): Firefox GUI Browser Verification

To earn **10 bonus points**, add a **5th node** to your topology: a Linux Desktop GUI Node featuring the **Firefox web browser** (<initials>-DESKTOP).

Challenge Requirements:

1. **Network & DNS Configuration:** Configure the Desktop node on the Client LAN (10.1.1.20/24), set its default gateway to Router <initials>-R1 (10.1.1.1), and set its DNS server to <initials>-DNS0 (10.1.2.53).
2. **Push Root of Trust into Firefox:** Transfer and import your Root CA certificate (ca.crt) directly into Firefox's internal Certificate Store (*Settings* → *Privacy & Security* → *Certificates* → *View Certificates* → *Authorities* → *Import...*).
3. **Browser HTTPS Verification:** Open Firefox and navigate to https://<lastname>.cs457.lab (e.g. https://haefner.cs457.lab).
4. **Deliverable:** Include a screenshot of the Firefox browser displaying your web page with a **trusted secure padlock icon** (confirming a valid, trusted SSL/TLS connection with zero security warnings!).

[!NOTE] Independent Configuration Challenge: Step-by-step terminal commands for this extra credit task are intentionally withheld—students must apply what they have learned to configure the desktop IP/gateway/DNS settings and Firefox certificate manager independently!

7. Wireshark Analysis and Questions

Open your packet capture in Wireshark. Filter by `dns || ssl || tls || tcp.port == 443` and answer the following questions:

Part A: DNS Resolution (UDP Port 53)

1. Locate the **DNS Query and Response** packets sent between Client (10.1.1.10) and DNS Server (10.1.2.53).
 - What is the packet number for the DNS Query for <lastname>.cs457.lab (e.g. haefner.cs457.lab)?
 - What IP address is returned in the DNS Answer?
 - What transport protocol and port number does DNS use for standard domain queries?

Part B: Inter-Subnet TCP Connection

2. Locate the **TCP 3-Way Handshake** between Client (10.1.1.10) and Web Server (10.1.2.100).
 - What are the packet numbers for SYN, SYN-ACK, and ACK?

- Explain how packets travel across subnets between `10.1.1.0/24` and `10.1.2.0/24` via default gateway `10.1.1.1` (`R1`).

Part C: PKI & Server Certificate Inspection

3. Locate the **TLS Certificate** message sent from Server to Client.
 - What is the **Common Name (CN)** of the Subject in your server certificate (e.g. `haefner.cs457.lab`)?
 - What is the **Common Name (CN)** of the Issuer CA in the server certificate (`CS457-Root-CA`)?
 - What digital signature algorithm was used by your private CA to sign the certificate?
 - What are the first four hexadecimal digits of the server's public key modulus?

Part D: TLS Handshake & Cipher Suite Negotiation

4. Inspect the **TLS Client Hello** message:
 - What version of TLS is declared by the client?
 - How many cipher suites were offered by the client?
 - What are the first two hexadecimal digits of the `Random Bytes` field? What is the purpose of this random nonce?
5. Inspect the **TLS Server Hello** message:
 - Which specific cipher suite was chosen by the server?
 - What symmetric encryption algorithm and key length does this cipher suite use?

Part E: Root of Trust Verification & Application Data

6. Locate the **Application Data** packets following the handshake:
 - Why are the HTTP GET request and HTTP response payloads encrypted and unreadable in Wireshark?
 - Explain how installing `ca.crt` on the client enabled `curl` to verify the server's identity and establish trust without security warnings.

8. Troubleshooting

Issue Encountered: [Describe any issues with routing, DNS resolution, OpenSSL, Nginx, or curl verification]

Discovery Method: [How did you identify the issue?]

Resolution: [How did you fix it?]

9. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- The Wireshark packet capture file (`tls_dns_traffic.pcap` or `tls_dns_traffic.pcapng`) containing the captured DNS queries, TCP handshake, and TLS session traffic.
- A single PDF report containing:
 - Your written answers to all 6 Wireshark analysis questions in Section 7.
 - Screenshot 1: Active CML topology showing running Client (`<initials>-H0`), Router (`<initials>-R1`), DNS (`<initials>-DNS0`), and Web Server (`<initials>-WEB0`).
 - Screenshot 2: DNS Server console showing `dnsmasq` running and successful `nslookup <lastname>.cs457.lab` (e.g. `haefner.cs457.lab`) on Client.
 - Screenshot 3: Web Server console showing successful creation of `ca.crt` , `server.key` , and `server.crt` using OpenSSL.
 - Screenshot 4: Client console showing successful `curl -v --cacert /tmp/ca.crt https://<lastname>.cs457.lab` output confirming `SSL certificate verify ok.` .

- Screenshot 5: Wireshark capture window showing DNS query/response, TCP 3-way handshake, and TLS Handshake / Application Data.
- **Optional Extra Credit Screenshot (+10 Points):** Screenshot of Firefox browser on Desktop node `<initials>-DESKTOP` displaying `https://<lastname>.cs457.lab` with a fully trusted secure padlock icon.